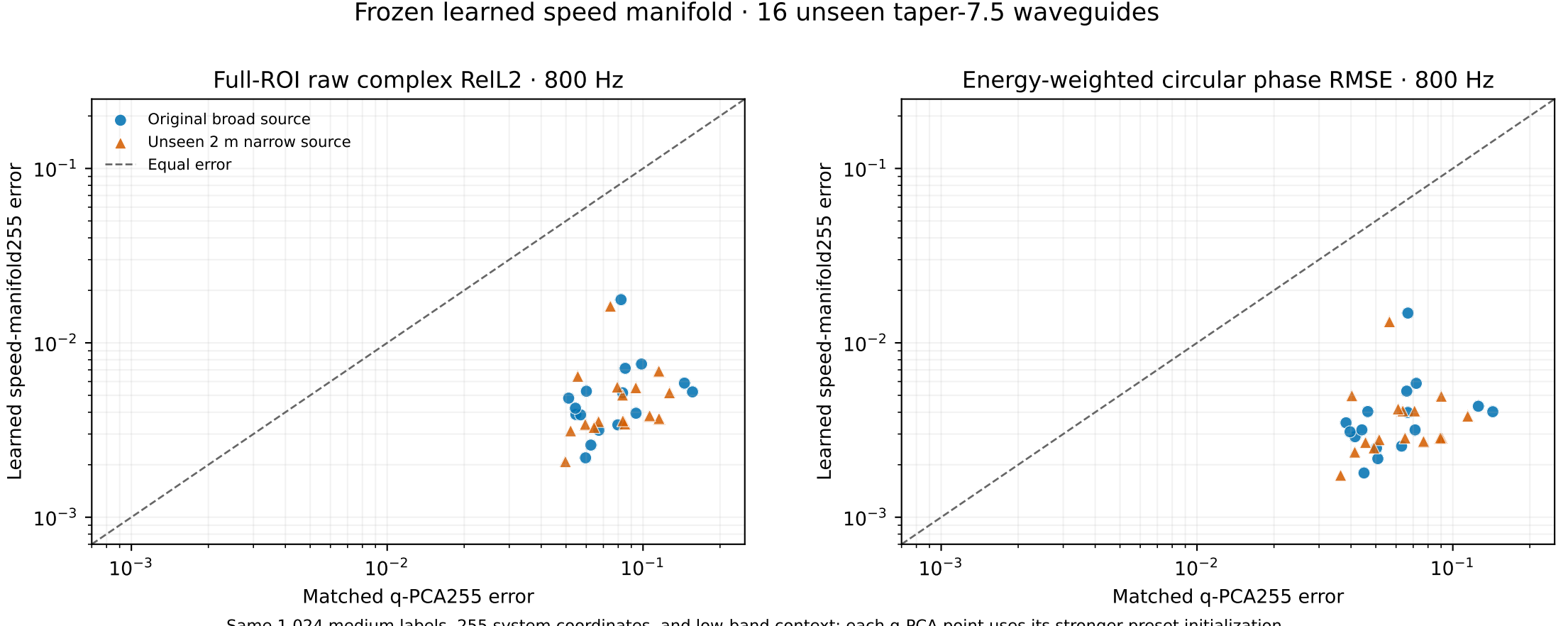
Frozen learned speed manifold · 16 unseen taper-7.5 waveguides
Full-ROI raw complex RelL2 · 800 Hz
Energy-weighted circular phase RMSE · 800 Hz
Original broad source
Unseen 2 m narrow source
Equal error
Learned speed-manifold255 error
Matched q-PCA255 error
10^{-1}
10^{-2}
10^{-3}
Same 1,024 medium labels, 255 system coordinates, and low-band context; each q-PCA point uses its stronger preset initialization.
Synthetic one-way waveguide family only; this is not a second physics domain.